

Foundations of Artificial Intelligence

Part VIII: AI in Society, Ethics & Emerging Trends

Department of Computer Science

Lecture Roadmap

1. Bias in AI Systems
2. Privacy Concerns
3. AI and Employment
4. Responsible AI
5. AI Governance
6. Generative AI & Large Language Models
7. AI for Social Good

Learning Objectives

By the end of this lecture, you should be able to:

- Identify and explain sources of bias in AI systems.
- Discuss privacy implications of data-driven AI.
- Analyse AI's impact on labour and employment.
- Articulate principles of Responsible AI and governance frameworks.
- Describe Generative AI and Large Language Models at a conceptual level.
- Recognise opportunities for AI to create social good, including agriculture and public policy.

Bias in AI Systems

Where Bias Comes From

Bias in AI arises at multiple stages:

- **Data collection:** datasets under-represent certain groups.
- **Label quality:** annotators bring implicit assumptions.
- **Model choice:** some architectures amplify existing patterns.
- **Deployment context:** model used outside the population it was trained on.

Core insight (from R&N, Ch. 26): An AI system optimises for measurable proxies; if those proxies are biased, the system inherits—and often amplifies—that bias.

Types of Bias

Type	Description
Historical bias	Patterns in data reflect past discrimination.
Representation bias	Certain groups appear less in training data.
Measurement bias	Features poorly measured for some sub-groups.
Aggregation bias	One model applied to diverse populations.
Deployment bias	System used in context different from design intent.

Example: Facial recognition systems with higher error rates on darker skin tones (Buolamwini & Gebru, 2018).

Measuring Fairness

Multiple mathematical definitions of fairness:

- **Demographic parity:** equal positive prediction rates across groups.
- **Equal opportunity:** equal true positive rates across groups.
- **Calibration:** predicted probabilities match actual outcomes equally well.

Key result: It is mathematically *impossible* to satisfy all definitions simultaneously when base rates differ across groups (Chouldechova, 2017).

Choosing a fairness criterion is therefore an **ethical and policy decision**, not just a technical one.

Mitigating Bias

Three intervention points:

1. **Pre-processing:** re-sample, re-weight, or augment training data.
2. **In-processing:** add fairness constraints to the learning objective.
3. **Post-processing:** adjust decision thresholds per group after training.

Complementary practices:

- Diverse teams building diverse datasets.
- Regular bias audits throughout the model lifecycle.
- Participatory design involving affected communities.

Privacy Concerns

Modern AI is data hungry:

- Training on web-scale corpora may include personal information.
- Inference can reveal sensitive attributes not explicitly shared.
- Models can memorise and reproduce training samples.

From the textbook (Ch. 26): “Computerization will lead to loss of privacy.”

Privacy risk categories:

- Direct identification from outputs.
- Re-identification by combining predictions with external data.
- Membership inference attacks.

Privacy-Preserving Techniques

- **Differential privacy:** add calibrated noise so individual records cannot be inferred.
- **Federated learning:** train on distributed data; model updates (not raw data) are shared.
- **Data minimisation:** collect and retain only what is strictly necessary.
- **Anonymisation:** remove identifiers before training (but beware re-identification).

Regulatory context: GDPR (EU), CCPA (California), and emerging national frameworks set binding requirements on data use and consent.

AI and Employment

The Automation Question

AI and automation historically:

- Displace **routine, repetitive** tasks (both cognitive and physical).
- Create new roles requiring human judgment, creativity, and adaptability.

Recent evidence:

- Tasks structured around rules are most vulnerable.
- Jobs combining physical dexterity + social intelligence are harder to automate.
- Generative AI now extends reach to knowledge work and content creation.

R&N perspective: AI creates wealth but its distribution is not guaranteed to be equitable.

AI's Impact on the Labour Market

Effect	Description	Example
Displacement	Tasks automated, jobs eliminated	Data entry, basic translation
Augmentation	Human + AI outperforms either alone	Radiology + AI screening
Creation	New roles emerge around AI systems	Prompt engineering, AI auditing
Polarisation	Demand shifts to high- and low-skill extremes	Middle-skill roles most at risk

Outcome depends on transition support, education policy, and regulation.

Adapting to AI-Driven Change

Strategies for society and individuals:

- **Continuous upskilling:** lifelong learning and digital literacy.
- **Education reform:** prioritise critical thinking, creativity, and interdisciplinary skills.
- **Policy instruments:** social safety nets, portable benefits, retraining programmes.
- **Complementarity:** design AI tools that amplify human capabilities.

Responsible AI

What Is Responsible AI?

Responsible AI is a set of **principles and practices** to ensure AI systems are:

- **Fair:** do not discriminate unjustly.
- **Reliable & Safe:** behave correctly under normal and edge-case conditions.
- **Transparent:** decisions are explainable to users and regulators.
- **Privacy-preserving:** handle personal data with care.
- **Inclusive:** accessible to and beneficial for diverse populations.
- **Accountable:** humans can identify and correct failures.

Explainability and Interpretability

Interpretable model: inherently understandable (e.g., decision tree, logistic regression).

Explainable model: post-hoc explanations for black-box models.

Explanation methods:

- **LIME:** locally approximate model behaviour around a specific prediction.
- **SHAP:** attribute prediction contribution to each input feature.
- **Attention maps:** visual evidence for image/text model decisions.
- **Counterfactuals:** “If feature X had been different, the outcome would have changed.”

Explainability is not optional in **high-stakes domains**: healthcare, criminal justice, credit scoring.

Responsibility and Accountability

A challenge highlighted in foundational AI literature:

*“One particular risk is **loss of accountability**: if an AI system makes a bad decision, who is responsible?”*

Stakeholders involved:

- Data providers — responsible for data quality and consent.
- Developers — responsible for model design and testing.
- Deployers — responsible for context and operational safeguards.
- Regulators — responsible for setting standards and enforcement.
- Users — responsible for appropriate use.

AI Governance

Why AI Governance?

Without governance, AI systems can:

- Perpetuate or amplify social inequalities.
- Operate in domains requiring oversight without adequate checks.
- Pose safety risks in high-stakes environments.
- Concentrate economic and political power.

Governance brings:

- Rules and standards that constrain harmful deployment.
- Incentives for responsible development.
- Redress mechanisms for affected individuals.

Governance Frameworks (Global Overview)

Framework / Region	Key Approach
EU AI Act (2024)	Risk-based tiering; high-risk systems face strict requirements; bans some uses.
OECD AI Principles	Human-centred values, transparency, robustness, accountability, inclusiveness.
US Executive Order (2023)	Safety testing, watermarking generative AI, federal agency guidance.
UNESCO AI Ethics	Global recommendation on data governance, environment, education, and gender.
National frameworks	Countries developing their own policies; divergence in approach is a challenge.

Technical Governance Tools

- **Model cards:** document model purpose, performance across sub-groups, and intended use.
- **Datasheets for datasets:** record provenance, collection methods, and known limitations.
- **Algorithmic impact assessments:** structured process to evaluate potential harms before deployment.
- **Audit trails and logging:** maintain records of model decisions for later review.
- **Red-teaming:** adversarial testing by independent teams to find safety issues.

Generative AI & Large Language Models

What Is Generative AI?

Generative AI produces **new content** (text, images, code, audio, video) rather than only classifying existing content.

Key model families:

- **Autoregressive models (GPT):** predict next token given context.
- **Diffusion models:** learn to reverse a noise process to generate images.
- **VAEs and GANs:** encode and decode latent representations.

Common applications: writing assistance, code generation, image synthesis, music, and video.

Large Language Models (LLMs)

LLMs are autoregressive transformers trained on internet-scale text.

Key capability: Predict

$$P(\text{token}_t \mid \text{token}_{1:t-1})$$

across billions of parameters, learning grammar, facts, and reasoning patterns.

Emergent abilities at scale:

- Few-shot and zero-shot task completion.
- Step-by-step reasoning (chain-of-thought prompting).
- Code generation, translation, and summarisation.

How LLMs Work (Conceptual)

Three-stage development cycle:

1. **Pre-training:** self-supervised next-token prediction on large corpora.
2. **Fine-tuning (SFT):** supervised training on curated instruction-response pairs.
3. **Alignment (RLHF):** human feedback used to improve helpfulness and safety.

Transformer architecture:

- Self-attention: each token attends to all others in context.
- Feed-forward layers: apply non-linear transformations.
- Scales well with data and compute.

Capabilities and Limitations of LLMs

Capabilities:

- Broad knowledge across domains from training data.
- Flexible generation: adapt tone, style, and format on request.
- Strong in-context learning without retraining.

Limitations:

- **Hallucination:** confidently generate false information.
- **No real-time knowledge:** knowledge cut-off from training.
- **Bias amplification:** reproduce and magnify biases in training data.
- **Misuse risk:** disinformation, phishing, manipulation at scale.
- **Computational cost:** training and inference are energy-intensive.

AI for Social Good

AI creates opportunities to address large-scale societal challenges:

- **Healthcare:** disease detection, drug discovery, diagnostic support.
- **Education:** personalised learning, accessibility tools, tutoring.
- **Environment:** climate modelling, wildlife conservation, energy optimisation.
- **Disaster response:** satellite imagery analysis, resource coordination.
- **Legal/civic:** document summarisation, access to information.

Success depends on: **quality data**, **community involvement**, and **sustainable deployment**.

AI in Agriculture

Agriculture accounts for a significant share of employment and food security in many countries.

How AI can help:

- **Crop disease detection:** image classification on leaf photos to diagnose disease early.
- **Yield prediction:** regression models using weather, soil, and satellite data.
- **Precision irrigation:** sensor data + ML to water only where and when needed.
- **Pest and weed management:** targeted treatment reduces chemical use.
- **Market price prediction:** helps smallholder farmers plan and reduce losses.

Mobile-first, low-bandwidth models are critical for low-resource settings.

Governments are adopting AI across policy domains:

- **Public health:** epidemiological modelling, outbreak detection, resource allocation.
- **Transport:** traffic prediction, public transit optimisation.
- **Social services:** fraud detection in benefit systems, eligibility triage.
- **Urban planning:** predictive analysis for infrastructure and housing needs.
- **Tax and revenue:** anomaly detection in filings.

Risks: automated decisions affecting citizens require transparency, appeal mechanisms, and human oversight.

Power and Responsibility

AI systems concentrates significant power:

- Economic: platform lock-in, IP concentration.
- Political: surveillance capability, information shaping.
- Social: algorithmic gatekeeping of opportunity.

Countervailing forces:

- Open-source models reduce barriers to access.
- Civil society audits and advocacy.
- International coordination on standards.
- Education and public AI literacy.

Core principle: The greater the capability, the greater the obligation to deploy responsibly.

Live Demo (In-Class Activity)

Interactive file: `ai-ethics-governance-demo.html`

Suggested walkthrough (10 min):

1. **Bias and Fairness:** adjust group outcomes and interpret parity gap.
2. **Privacy Risk:** vary noise and retention to show utility-risk trade-off.
3. **Employment Impact:** compare displacement pressure vs augmentation.
4. **Governance Check:** run checklist to decide deployment readiness.
5. **Generative AI Risk:** scan prompts for policy and misuse signals.

Open locally in browser: Lecture

`8/ai-ethics-governance-demo.html`

Summary and Course Conclusion

What we covered today:

- Bias in AI systems and fairness trade-offs.
- Privacy risks and mitigation techniques.
- AI's nuanced effects on employment.
- Responsible AI principles and explainability.
- AI governance frameworks and tools.
- Generative AI and LLMs: power and pitfalls.
- Social good applications: agriculture and public policy.

Key takeaway: Technical capability must always be paired with ethical reflection, inclusive design, and accountable governance.

“The question is not whether we will have AI—it is what kind of AI we choose to build, and for whom.”